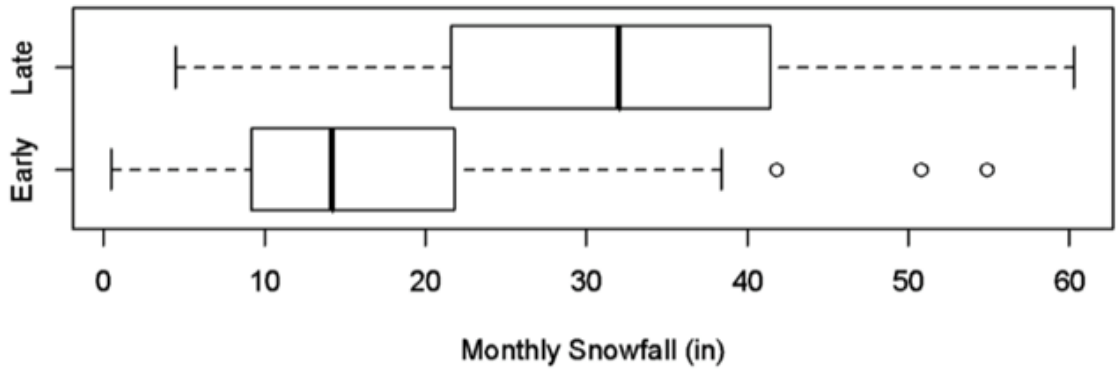
Circle the correct answer for each question. If you're unsure, you can write numbers 1, 2, 3, etc. next to the choices to get partial credit. Questions refer to the nonindented paragraph above unless otherwise stated.

Question permutation table:

Version		1	2	3	4
Question					
1	→	2	1	2	1
2*	→	3	2	3	2
3*	→	4	3	4	3
4*	→	5	4	5	4
5*	→	6	5	6	5
6	→	1	6	1	6

*: Question follows previous

Monthly snowfall (in inches) was measured over several winters in Fort Collins, Colorado. Researchers also recorded whether the measurement was taken in the Early winter (September to December) or Late winter (January to June). Boxplots displaying the distribution of monthly snowfall for each season are below.



Use this to answer the next five questions:

1. The shape of the distribution of monthly snowfall measurements for the Early season is

   (a) symmetric.
   (b) skewed right.
   (c) skewed left.
   (d) bimodal.

Correct answers: (c)

Answer permutations: (All answers permuted)

Version		1	2	3	4
Answer					
(a)	→	(d)	(c)	(a)	(d)
(b)	→	(c)	(b)	(c)	(a)
(c)	→	(a)	(d)	(d)	(b)
(d)	→	(b)	(a)	(b)	(c)

2. What is the upper quartile (Q3) of the distribution of monthly snowfall measurements for the Late season (approximately)?

   (a) 22
   (b) 33
   (c) 41
   (d) 60

   Correct answers: (c)

   Answer permutations: (All answers permuted)

	Version	1	2	3	4
Answer					
(a)	→	(a)	(d)	(c)	(d)
(b)	→	(d)	(c)	(b)	(a)
(c)	→	(c)	(a)	(a)	(b)
(d)	→	(b)	(b)	(d)	(c)

3. Which of the following sentences correctly interprets the first quartile (Q1) of the distribution of monthly snowfall measurements for the Early season?

   (a) About 25% of Early season winter months in Fort Collins get less than 9 inches of snow.
   (b) About 50% of Early season winter months in Fort Collins get less than 9 inches of snow.
   (c) About 75% of Early season winter months in Fort Collins get less than 9 inches of snow.
   (d) About 25% of Early season winter months in Fort Collins get more than 38 inches of snow.

   Correct answers: (a)

   Answer permutations: (All answers permuted)

	Version	1	2	3	4
Answer					
(a)	→	(a)	(b)	(a)	(d)
(b)	→	(c)	(d)	(b)	(a)
(c)	→	(d)	(a)	(d)	(c)
(d)	→	(b)	(c)	(c)	(b)

4. Which season has the larger median monthly snowfall?

   (a) Early
   (b) Late
   (c) The two medians are approximately equal.
   (d) The plot does not provide enough information to determine which median is larger.

   Correct answers: (b)

   Answer permutations: (All answers permuted)

	Version	1	2	3	4
Answer					
(a)	→	(b)	(b)	(b)	(a)
(b)	→	(d)	(c)	(d)	(d)
(c)	→	(a)	(d)	(a)	(b)
(d)	→	(c)	(a)	(c)	(c)

5. The boxplots demonstrate that there is an association between which two variables?

   (a) Early and Late winter season
   (b) Snowfall and months
   (c) Monthly snowfall and whether the time was Early or Late winter season
   (d) Whether it was Early or Late winter season and months

   Correct answers: (c)

Answer permutations: (All answers permuted)

	Version	1	2	3	4
Answer					
(a)	→	(a)	(d)	(b)	(d)
(b)	→	(b)	(b)	(c)	(a)
(c)	→	(c)	(a)	(a)	(b)
(d)	→	(d)	(c)	(d)	(c)

6. Which of the following data sets has the largest standard deviation?

   (a) 1, 2, 3, 4, 5
   (b) 1, 3, 3, 3, 5
   (c) 1, 1, 3, 5, 5
   (d) 1, 1, 1, 1, 1

Correct answers: (c)

Answer permutations: (All answers permuted)

	Version	1	2	3	4
Answer					
(a)	→	(a)	(a)	(c)	(a)
(b)	→	(d)	(c)	(d)	(d)
(c)	→	(c)	(b)	(b)	(c)
(d)	→	(b)	(d)	(a)	(b)